

Statistics & Data Analysis Assignment — **tips** Dataset

Part 1 — Get to Know the Data

Task 3

For each column, identify whether it is:

- Categorical
- Numerical

Task 4

Which variables describe the customer?

Task 5

Which variables describe the bill?

Task 6

Which variables describe the tip?

Task 7

Which variables could be considered potential factors related to the tip amount?

Part 2 — Frequency & Basic Exploration

Task 8

How many customers are smokers and how many are non-smokers?

Task 9

How many customers belong to each gender?

Task 10

How many customers visited the restaurant on each day?

Task 11

Which day has the largest number of recorded customers?

Task 12

How many customers visited during lunch and how many during dinner?

Task 13

Which party size occurs most frequently?

Task 14

Create an appropriate visualization showing the number of customers for each day.

Task 15

Create an appropriate visualization showing the distribution of customers between lunch and dinner.

Part 3 — Central Tendency

Focus on the numerical variables:

- `total_bill`
- `tip`
- `size`

Task 16

What is the average total bill?

Task 17

What is the median total bill?

Task 18

What is the most frequent party size?

Task 19

What is the average tip?

Task 20

What is the median tip?

Task 21

Compare the mean and median of `total_bill`.

Task 22

Compare the mean and median of `tip`.

Task 23

For which variable is the mean noticeably different from the median?

Task 24

What might the difference between the mean and median tell you about the data?

Part 4 — Measuring Spread

Now investigate whether the average gives a complete picture of the data.

Task 25

What is the minimum and maximum total bill?

Task 26

What is the range of the total bill?

Task 27

What is the minimum and maximum tip?

Task 28

What is the range of the tip?

Task 29

Calculate the variance of `total_bill`.

Task 30

Calculate the standard deviation of `total_bill`.

Task 31

Calculate the variance of `tip`.

Task 32

Calculate the standard deviation of `tip`.

Task 33

Which variable has greater variability: `total_bill` or `tip`?

Task 34

Why is standard deviation easier to interpret than variance in the original units of the data?

Task 35

If two groups of customers have the same average bill but different standard deviations, what does that tell you about the two groups?

Part 5 — Quartiles, IQR & Outliers

Task 36

Find Q1, Q2, and Q3 for `total_bill`.

Task 37

Find the interquartile range of `total_bill`.

Task 38

Find Q1, Q2, and Q3 for `tip`.

Task 39

Find the interquartile range of `tip`.

Task 40

What range contains the middle 50% of the total bills?

Task 41

What range contains the middle 50% of the tips?

Task 42

Create a boxplot for `total_bill`.

Task 43

Create a boxplot for `tip`.

Task 44

Are there any potential outliers in `total_bill`?

Task 45

Are there any potential outliers in `tip`?

Task 46

How might unusually large bills affect the mean total bill?

Part 6 — Distribution & Skewness

Now move from individual summary numbers to the **shape of the data**.

Task 47

Create a histogram for `total_bill`.

Task 48

Create a histogram for `tip`.

Task 49

Describe the shape of the `total_bill` distribution.

Task 50

Describe the shape of the `tip` distribution.

Task 51

Does the `total_bill` distribution appear approximately symmetric or skewed?

Task 52

Does the `tip` distribution appear approximately symmetric or skewed?

Task 53

Calculate the skewness of `total_bill`.

Task 54

Calculate the skewness of `tip`.

Task 55

Does the calculated skewness support what you observed in the histograms?

Task 56

If the mean is greater than the median, what does this suggest about the shape of the distribution?

Part 7 — Comparing Groups

Now investigate whether different customer groups behave differently.

Task 57

What is the average total bill for each day?

Task 58

What is the average tip for each day?

Task 59

Which day has the highest average total bill?

Task 60

Which day has the highest average tip?

Task 61

Compare the average tip between smokers and non-smokers.

Task 62

Compare the average total bill between smokers and non-smokers.

Task 63

Compare the average tip between male and female customers.

Task 64

Compare the average tip between lunch and dinner customers.

Task 65

Which grouping variable appears to show the largest difference in average tip?

Part 8 — Covariance

Now move from analyzing **one numerical variable** to studying **two numerical variables**.

Focus on:

`total_bill` and `tip`.

Task 66

Before calculating anything, do you expect customers with larger bills to generally leave larger tips?

Task 67

Calculate the covariance between `total_bill` and `tip`.

Task 68

Is the covariance positive or negative?

Task 69

What does the direction of the covariance tell you about the two variables?

Task 70

Why is it difficult to compare the strength of relationships using covariance alone?

Part 9 — Correlation

Task 71

Calculate the correlation coefficient between `total_bill` and `tip`.

Task 72

Is the relationship positive or negative?

Task 73

How strong is the relationship between total bill and tip?

Task 74

Create a scatter plot of `total_bill` against `tip`.

Task 75

Does the scatter plot support your correlation result?

Task 76

Are there any observations that appear unusual compared with the general pattern?

Task 77

Would you describe the relationship as weak, moderate, or strong?

Task 78

Does the correlation prove that a larger bill causes a larger tip? Explain.

Part 10 — Linear Regression

Now ask the business question:

Can we use the total bill to estimate the tip for a new customer?

Task 79

Which variable should be the independent variable?

Task 80

Which variable should be the dependent variable?

Task 81

Create a scatter plot of `total_bill` and `tip`.

Task 82

Add a linear trendline to the scatter plot.

Task 83

Display the regression equation on the chart.

Task 84

Display the R^2 value on the chart.

Task 85

Interpret the slope of the regression equation in the context of restaurant tips.

Task 86

Interpret the intercept of the regression equation.

Task 87

What does the R^2 value tell you about the ability of total bill to explain variation in tip?

Task 88

Use the regression equation to estimate the tip for a customer with a total bill of \$30.

Task 89

Estimate the tip for a customer with a total bill of \$50.

Task 90

Would you trust the regression model equally for every possible bill amount? Why?

Part 11 — The Statistical Twist: Sample vs Population

Now imagine:

The `tips` dataset represents only a **sample** of all restaurant customers.

Task 91

What is the population in this scenario?

Task 92

What is the sample?

Task 93

Is the average tip calculated from the dataset necessarily equal to the average tip of the entire population?

Task 94

Why might the sample mean be different from the population mean?

Task 95

If you could collect data from every customer, would you still need to estimate the population mean?

Part 12 — Sampling Experiment

Now investigate what happens when we repeatedly take samples.

Task 96

Take a random sample of 30 observations from the `tips` dataset.

Task 97

Calculate the mean tip of your first sample.

Task 98

Repeat the sampling process several times.

Task 99

Record the mean tip for each sample.

Task 100

Create a table containing the means of your repeated samples.

Task 101

Create a histogram of the sample means.

Task 102

Compare the distribution of the original **tip** values with the distribution of the sample means.

Task 103

Does the distribution of sample means have the same shape as the original **tip** distribution?

Task 104

What happens to the sample means when you repeat the sampling process?

Part 13 — Normal Distribution

Now investigate the question:

Why do statisticians often work with the normal distribution?

Task 105

What does a normal distribution look like?

Task 106

What characteristics distinguish a normal distribution from a strongly skewed distribution?

Task 107

Is the `tip` variable approximately normally distributed?

Task 108

Is the `total_bill` variable approximately normally distributed?

Task 109

Does every real-world dataset have to follow a normal distribution?

Task 110

Give an example of a variable that might reasonably follow an approximately normal distribution.

Task 111

Give an example of a variable that might not follow a normal distribution.

Part 14 — Standard Deviation & the 68–95–99.7 Rule

Assume a numerical variable follows an approximately normal distribution.

Task 112

If the mean is 20 and the standard deviation is 4, approximately what percentage of observations would be expected to fall within one standard deviation of the mean?

Task 113

Approximately what percentage would be expected within two standard deviations?

Task 114

Approximately what percentage would be expected within three standard deviations?

Task 115

For the same distribution, between which two values would approximately 68% of observations fall?

Task 116

Between which two values would approximately 95% of observations fall?

Task 117

Why can these rules not automatically be applied to every dataset?

Part 15 — Sampling Distribution

Now return to the repeated sampling experiment.

Task 118

What is a sampling distribution?

Task 119

How is a sampling distribution different from the distribution of the original observations?

Task 120

What statistic did you calculate for each sample in your experiment?

Task 121

What happens to the distribution of sample means when the number of repeated samples increases?

Task 122

What happens to the distribution of sample means when the sample size increases?

Part 16 — Central Limit Theorem

Task 123

Using your sampling experiment, investigate what happens to the distribution of sample means when the sample size is increased from 10 to 30.

Task 124

Repeat the experiment using a larger sample size.

Task 125

Compare the histograms of the sample means for the different sample sizes.

Task 126

Does the distribution of sample means become more approximately normal as the sample size increases?

Task 127

Does the original **tip** distribution need to be perfectly normal for the sampling distribution of the mean to become approximately normal?

Task 128

In your own words, describe the Central Limit Theorem.

Task 129

Why is the Central Limit Theorem useful when we only have a sample but want to learn about a population?

Part 17 — Confidence Interval: From Sample to Population

Now return to the original question:

What can we say about the average tip of the entire population?

Task 130

Calculate the sample mean of `tip`.

Task 131

Calculate the sample standard deviation of `tip`.

Task 132

Calculate a 95% confidence interval for the population mean tip.

Task 133

What are the lower and upper limits of the confidence interval?

Task 134

Interpret the confidence interval in the context of restaurant customers.

Task 135

Why is it better to report a confidence interval instead of claiming that the sample mean is exactly equal to the population mean?

Task 136

What would happen to the confidence interval if you increased the sample size?

Task 137

What would happen to the confidence interval if the data had greater variability?

Final Challenge — Tell the Story

Task 138

Imagine you are a **data analyst for the restaurant**.

Prepare a one-page statistical summary answering:

1. What is a typical customer bill?
2. What is a typical tip?
3. How much do bills and tips vary?
4. What does the distribution of tips look like?
5. Are bills and tips related?
6. How strong is their relationship?
7. Can total bill be used to predict tip?
8. What does the regression model tell the restaurant?
9. If this dataset is only a sample, what can we say about the larger population?
10. How does the Central Limit Theorem help us move from the sample to population-level inference?

Final question

What is the difference between “describing the data we have” and “making an inference about the population from which the data came”?